4. GRAVITATIONAL WAVES

i) Black holes will merge at this distance.

$r_{\min} = r_{s1} + r_{s2} = \frac{2G(m_1+m_2)}{c^2}$

Substituting for r in the power equation, we get the maximum power for a system with masses m_1 and m_2.

$P_{\max}(m_1, m_2) = \frac{1}{5}\frac{c^5}{G}\frac{(m_1 m_2)^2}{(m_1+m_2)^4}$

Now let us consider the effect of masses. Let us define $a := \frac{(m_1 m_2)^2}{(m_1+m_2)^4}$. The intuition tells us, that more power should be radiated off a bigger system, thus the masses should tend to infinity for maximum power.

However the limit $\lim_{m_1=m_2\to\infty} a = 1$ suggests otherwise. So does the dimensional analysis: the dimension of $[a] = \frac{(M\cdot M)^2}{M^4} = 1$ means that the absolute values of the masses don't contribute to the equation. This is intuitively explained by the proportionally increasing Schwarzschild radii.

However the mass distribution does matter, since a isn't the same for all m_1, m_2 either. Let us determine the maximum value it can

take. Let us redefine it in terms of the total mass $M := m_1 + m_2$ and the mass distribution $k := \frac{m_1}{M}$. Then

$m_1 = kM$

$m_2 = (1-k)M$

$\Rightarrow a = \frac{(m_1 m_2)^2}{(m_1+m_2)^4} = \frac{(kM(k-1)M)^2}{M^4} = [k(k-1)]^2$

$\frac{d}{dk}[k(k-1)]^2 = 0 \Leftrightarrow k = \frac{1}{2} \Leftrightarrow m_1 = m_2$

$\Rightarrow a_{\max} = [\frac{1}{2}(1-\frac{1}{2})]^2 = \frac{1}{16}$

So that finally

$P_{\max} = \frac{1}{5}\frac{c^5}{G}\frac{1}{16} = \frac{1}{80}\frac{c^5}{G} = 4.5 \times 10^{50}\,\mathrm{W}$

Interestingly enough, the peak power emitted in gravitational radiation by any merging binary system will be the same. $\frac{c^5}{G}$ is also called luminosity of the universe.

ii) $U = \frac{1}{2}k\Delta\ell^2$, energy stored in a uniformly deformed body

$k = \frac{F}{\Delta\ell}$

$F = \sigma S$

$\Delta\ell = \varepsilon\ell$

$\sigma = \varepsilon E$

$u = \frac{U}{V}$, where $V = S\ell$ is the volume

$\Rightarrow u = \frac{U}{V_0} = \frac{1}{2}E\varepsilon^2$

iii) $[G] = L^3M^{-1}T^{-2}$, from e.g. $F = \frac{GMm}{r^2}$

$[c] = LT^{-1}$

$[f] = T^-1$

$[E] = ML^{-1}T^{-2}$, from e.g. $\sigma = \varepsilon E$

let $E = G^\alpha c^\beta f^\gamma$. Solving for each dimension yields

$M: 1 = -\alpha \quad \Rightarrow \alpha = -1$

$L: -1 = 3\alpha + \beta \quad \Rightarrow \beta = 2$

$T: -2 = -2\alpha - \beta - \gamma \quad \Rightarrow \gamma = 2$

$\Rightarrow E = \frac{c^2}{G}f^2$

iv) The gravitational wave spreads as a spherical wave centered at the source. It's wavefront occupies a thin spherical shell with the surface area $S = 4\pi z^2$ and its associated infinitesimal volume is $dV = S\,dz$, where $dz = c\,dt$ for a wave traveling with the speed of light c. The power P and energy density u at a distance z is coupled through $dE = P\,dt = u\,dV \Leftrightarrow z = \sqrt{\frac{P}{4\pi u c}}$.

Using the models from

$$P = \frac{1}{80}\frac{c^5}{G}$$
$$u = \frac{U}{V_0} = \frac{1}{2}E\varepsilon^2$$
$$E = \frac{c^2}{G}f^2$$
we finally get
$$z = \frac{1}{\sqrt{160\pi}}\frac{c}{\varepsilon f}$$